

PHYSICS, CHEMISTRY & MATHEMATICS

PT - I

13.08.2023

Paper – 1

Time Allotted: 3 Hours

Maximum Marks: 228

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Sections.
3. **Section-I** is Physics, **Section -II** is Chemistry and **Section -III** is Mathematics.
4. Each section is further divided into Two Parts: **Part-A & Part-B**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with a **blue / black ball point pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Sections.

- (i) **Part-A (1 – 8)** contains 8 multiple choice questions which have **only one** correct answer. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Part-A (9 – 14) contains 6 multiple choice questions which have **one or more than one** correct answer. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Part-A (15 – 18) contains 2 Paragraphs. Based upon each paragraph, 2 Multiple Choice Questions which have **only one** correct answer. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

- (ii) **Part –B (1 – 2)**, contains 2 Matrix Match Type Questions which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+8 marks** for all correct answer however for each correct row **+2 marks** will be awarded. There is **no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

Batch: - LKTYR235A03

USEFUL DATA

PHYSICS		CHEMISTRY	
Acceleration due to gravity	: $g = 10 \text{ m/s}^2$	Gas Constant	$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Planck constant	: $h = 6.6 \times 10^{-34} \text{ J-s}$		$= 0.0821 \text{ Lit atm}$
Charge of electron	: $e = 1.6 \times 10^{-19} \text{ C}$	$\text{K}^{-1} \text{ mol}^{-1}$	
Mass of electron	: $m_e = 9.1 \times 10^{-31} \text{ kg}$		$= 1.987 \approx 2 \text{ Cal}$
Permittivity of free space	: $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / \text{N-m}^2$	$\text{K}^{-1} \text{ mol}^{-1}$	
Density of water	: $\rho_{\text{water}} = 10^3 \text{ kg/m}^3$	Avogadro's Number N_a	$= 6.023 \times 10^{23}$
Atmospheric pressure	: $P_a = 10^5 \text{ N/m}^2$	Planck's constant h	$= 6.625 \times 10^{-34} \text{ J.s}$
Gas constant	: $R = 8.314 \text{ J}$		$= 6.625 \times 10^{-27} \text{ erg.s}$
$\text{K}^{-1} \text{ mol}^{-1}$		1 Faraday	$= 96500 \text{ coulomb}$
		1 calorie	$= 4.2 \text{ joule}$
		1 amu	$= 1.66 \times 10^{-27} \text{ kg}$
		1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Atomic No:	H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9, Ne = 10, Na = 11, Mg = 12, Si = 14, Al = 13, P = 15, S = 16, Cl = 17, Ar = 18, K = 19, Ca = 20, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Zn = 30, As = 33, Br = 35, Ag = 47, Sn = 50, I = 53, Xe = 54, Ba = 56, Pb = 82, U = 92.		
Atomic masses:	H = 1, He = 4, Li = 7, Be = 9, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Si = 28, Al = 27, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Co = 59, Ni = 58.7, Cu = 63.5, Zn = 65.4, As = 75, Br = 80, Ag = 108, Sn = 118.7, I = 127, Xe = 131, Ba = 137, Pb = 207, U = 238.		

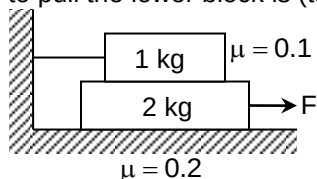
Section – I (Physics)

PART – A

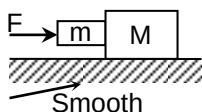
(Single Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

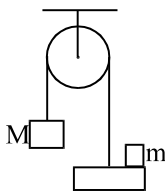
1. Minimum force required to pull the lower block is (take $g = 10 \text{ m/s}^2$)



- (A) 1 N (B) 5 N
(C) 7 N (D) 10 N
2. The blocks of masses m and M are not attached. The coefficient of static friction between the blocks is μ , but the surface beneath M is smooth. What is the maximum magnitude of the horizontal force F required to hold it against M :

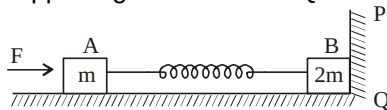


- (A) $\frac{m(M+m)g}{M}$ (B) $\frac{m(M+m)g}{\mu M}$
(C) $\frac{M(M+m)g}{\mu M}$ (D) $m(M+m)g$
3. Two identical weights of mass M are linked by a thread wrapped around a frictionless pulley with a fixed axis. A small weight of mass ' m ' is placed on one of the weights. What is reaction force between m and M ?

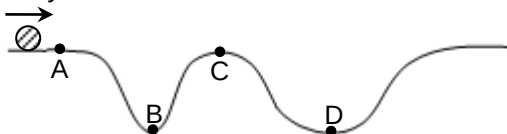


- (A) $\frac{2mMg}{2M+m}$ (B) $\frac{Mg(M+m)}{2M+m}$
(C) $\frac{mg(M+m)}{2M+m}$ (D) $\frac{2Mg(M+m)}{2M+m}$

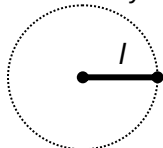
4. Figure shows a system of two blocks connected by a spring, kept on a frictionless surface. Block A is in equilibrium. Suddenly supporting vertical wall PQ is removed.



- (A) Initial acceleration of block A is $\frac{F}{m}$ (B) Initial acceleration of block B is $\frac{F}{2m}$
 (C) Initial acceleration of block B is zero (D) Initial acceleration of block A is $\frac{F}{3m}$
5. A body moves along an uneven road surface with constant speed at all points. The normal reaction of road on the body is



- (A) Maximum at A (B) Maximum at B
 (C) Maximum at C (D) None
6. A small sphere of mass m is suspended by a thread of length l . It is raised upto the height of suspension with thread fully stretched and released. Then the maximum tension in thread will be



- (A) mg (B) $2mg$
 (C) $3mg$ (D) $6mg$
7. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time as $a_c = k^2 r t^2$, where k is a constant. The power delivered to the particle by the forces acting on it is
- (A) $2\pi m k^2 r^2 t$ (B) $m k^2 r^2 t$ (C) $\frac{1}{3} m k^4 r^2 t^5$ (D) 0
8. A constant power P is applied to a particle of mass m . The distance travelled by the particle when its velocity increases from v_1 to v_2 is (neglect friction) :
- (A) $\frac{3P}{m}(v_2^2 - v_1^2)$ (B) $\frac{m}{3P}(v_2 - v_1)$ (C) $\frac{m}{3P}(v_2^3 - v_1^3)$ (D) $\frac{m}{3P}(v_2^2 - v_1^2)$

PART – A

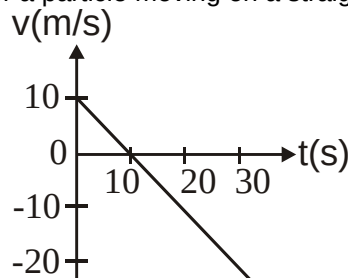
(Multi Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

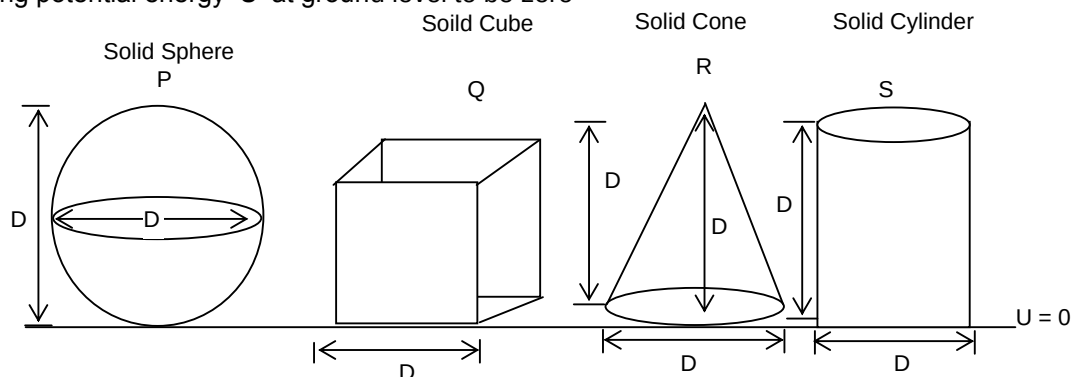
9. Given $\vec{A} = 3\hat{i} + 4\hat{j}$ & $\vec{B} = 6\hat{i} + 8\hat{j}$ which of the following statement is correct

- (A) $\vec{A} \times \vec{B} = 0$ (B) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$
 (C) $|\vec{A}| = 15$ (D) $\vec{A} \cdot \vec{B} = 48$

10. The velocity-time plot for a particle moving on a straight line is shown in fig.



- (A) The particle has constant acceleration
 (B) The particle has never turned around.
 (C) The particle has zero displacement
 (D) The average speed in the interval 0 to 10s is the same as the average speed in the interval 10s to 20s.
11. Assuming potential energy 'U' at ground level to be zero



All objects are made up of same material

U_P = Potential energy of solid sphere

U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone

U_S = Potential energy of solid cylinder

(A) $U_S > U_P$

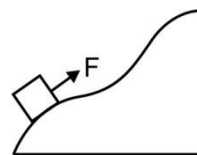
(B) $U_Q > U_S$

(C) $U_P > U_Q$

(D) $U_S > U_R$

12. A body of mass m was slowly hauled up the rough hill by a force F which at each point was directed along tangent to the hill. Work done by the force :

- (A) is independent of shape of trajectory
 (B) depends upon vertical components of displacement but independent of horizontal component
 (C) depends upon both the components of displacement
 (D) does not depend upon coefficient of friction



13. A block is suspended by an ideal spring of force constant k . If the block is pulled down by applying a constant force F and if maximum displacement of block from its initial position of rest is x_0 then

(A) increase in energy stored in spring is kx_0^2

(B) $x_0 = \frac{3F}{2k}$

(C) $x_0 = \frac{2F}{k}$

(D) work done by applied force F is Fx_0

14. A particle moves in a straight line with constant acceleration under a constant force F . Select the correct alternative(s)
- (A) power developed by this force varies linearly with time
 (B) power developed by this force varies parabolically with time
 (C) power developed by this force varies linearly with displacement
 (D) power developed by this force varies parabolically with displacement

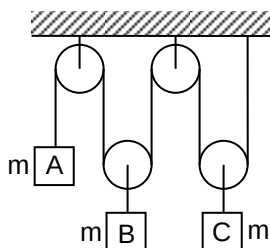
PART – A

(Comprehension Type)

This section contains **2 Paragraphs**. Based upon each paragraph, there are **2 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Comprehension 1:

Consider the system with all pulleys and threads are ideal. The mass of each block is m .



15. The acceleration of C is
- (A) $\frac{2g}{23}$ downwards (B) $\frac{4g}{7}$ upwards
 (C) $\frac{g}{4}$ upwards (D) $\frac{g}{9}$ upwards
16. Tension acting on block A is
- (A) $\frac{5}{9}mg$ (B) $\frac{25}{23}mg$
 (C) $\frac{3}{4}mg$ (D) $\frac{7}{9}mg$

Comprehension 2:

The scalar product of two vectors is defined as a scalar quantity equal to the product of their magnitudes and cosine of the angle between them.

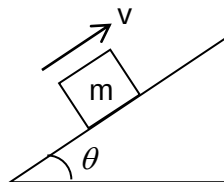
$$C = \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

Example Work done $W = \vec{F} \cdot \vec{S}$ and Power $P = \vec{F} \cdot \vec{V}$

Answer the following questions from above data.

17. A force $F = (2\hat{i} + 4\hat{j})\text{N}$ produces a displacement of $(3\hat{i} + \hat{j} - 4\hat{k})\text{m}$. Work done by force is
- (A) 10 J (B) 22 J
 (C) Zero (D) None of these

18. A block of mass m moving with a constant velocity v on a smooth inclined plane in upward direction under the action of an external force. The power generated by force is



- (A) $-mgv \sin \theta$ (B) $+mgv \sin \theta$
(C) $+mgv$ (D) Zero

PART – B

(Matrix Match Type)

This section contains **2 questions**. Each question has two columns. For each statement in Column I, there may be **ONE or MORE THAN ONE** correct choice(s) in Column II.

1. Column-I show vector diagram relating three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. Match the vector equation in column-II, with vector diagram in column-I :

Column I		Column II	
(A)		(p)	$\vec{a} - (\vec{b} + \vec{c}) = 0$
(B)		(q)	$\vec{b} - \vec{c} = \vec{a}$
(C)		(r)	$\vec{a} + \vec{b} = -\vec{c}$
(D)		(s)	$\vec{a} + \vec{b} = \vec{c}$

2. A single conservative force acts on a body of mass 1 kg that moves along the x-axis. The potential energy $U(x)$ is given by $U(x) = 20 + (x-2)^2$ where x is in meters. At $x = 5.0$ m the particle has a kinetic energy of 20 J then –

Column-I	Column-II
(A) minimum value of x in meters	(p) 29
(B) maximum value of x in meters	(q) 7.38
(C) maximum potential energy in joules	(r) 49
(D) maximum kinetic Energy in joules	(s) – 3.38

Section – II (Chemistry)

PART – A

(Single Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

- Which is not correctly matched?
 (A) XeO_3 : Trigonal bipyramidal
 (B) ClF_3 : T shaped
 (C) XeOF_4 : Square pyramidal
 (D) XeF_2 : Linear shape
- Which of the following species have the same bond order?
 (I) CO (II) CN^- (III) O_2^+
 (IV) NO^+
 (A) I, II, III (B) I, II, IV
 (C) I, III, IV (D) II, III, IV
- The electronic configurations of some neutral elements are given below -
 (1) $1s^2, 2s^2$ (2) $1s^2, 2s^2 2p^1$
 (3) $1s^2, 2s^1$ (4) $1s^2, 2s^2 2p^3$
 Which of these electronic configurations would be expected to have highest second ionisation energy (IE_2)?
 (A) (1) (B) (2)
 (C) (3) (D) (4)
- The correct order of increasing covalent character of the following is
 (A) $\text{SiCl}_4 < \text{AlCl}_3 < \text{CaCl}_2 < \text{KCl}$ (B) $\text{KCl} < \text{CaCl}_2 < \text{AlCl}_3 < \text{SiCl}_4$
 (C) $\text{AlCl}_3 < \text{CaCl}_2 < \text{KCl} < \text{SiCl}_4$ (D) none of these
- If molar ratio of Cl – 35 and Cl – 37 becomes 1 : 3, then what will be average atomic mass of natural chlorine?
 (A) 35.5 (B) 36.5
 (C) 37.5 (D) None of these
- According to Bohr's theory, the angular momentum of an electron in 5^{th} orbit is -
 (A) 25 h/p (B) 1.0 h/p
 (C) 10 h/p (D) 2.5 h/p
- The ratio of the energy of a photon of $2000 \text{ \AA}$ wavelength radiation to that of $4000 \text{ \AA}$ radiation is
 (A) $1/4$ (B) 4
 (C) $1/2$ (D) 2
- The correct order of second ionization potential of carbon nitrogen, oxygen and fluorines is
 (A) $\text{C} > \text{N} > \text{O} > \text{F}$ (B) $\text{O} > \text{N} > \text{F} > \text{C}$
 (C) $\text{O} > \text{F} > \text{N} > \text{C}$ (D) $\text{F} > \text{O} > \text{N} > \text{C}$

PART – A

(Multi Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- Which of the following molecules have odd electron?
 (A) NO (B) NO_2
 (C) ClO_2 (D) N_2O_4

10. Which of the following statements is true?
 (A) The dipole moment of NF_3 is zero
 (B) The dipole moment of NF_3 is less than NH_3
 (C) The dipole moment of NF_3 is more than NH_3
 (D) The dipole moment of NH_3 is non-zero
11. The species(s) having sp^3 hybridization is/are
 (A) H_2O (B) NH_3
 (C) NH_4^+ (D) NH_2^-
12. 8 g of oxygen has the same number of molecules as in :
 (A) 11 g of CO_2 (B) 22 g of CO_2
 (C) 7 g of CO (D) 14 g of CO
13. The ground state electronic configuration of nitrogen atom can be represented by
 (A) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{\uparrow} \boxed{\uparrow}$ (B) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{\downarrow} \boxed{\uparrow}$
 (C) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{\downarrow} \boxed{\downarrow}$ (D) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\downarrow} \boxed{\uparrow\downarrow} \boxed{\downarrow}$
14. In which of following configuration both Pauli and Hund's rule violate
 (A) $\boxed{\uparrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{}$ (B) $\boxed{\uparrow} \boxed{\uparrow\downarrow} \boxed{\downarrow} \boxed{\downarrow}$
 (C) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow} \boxed{}$ (D) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\downarrow}$

PART – A

(Comprehension Type)

This section contains **2 Paragraphs**. Based upon each paragraph, there are **2 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Comprehension 1:

MOT helps in determining B.O., B.L., magnetic behaviour of compounds. If a species has an unpaired electron in its molecular orbital, it shows paramagnetism and $\text{B.O.} = \frac{N_b - N_a}{2}$.

15. Which sequence correctly describes a relative bond strength, of oxygen molecule, superoxide ion and peroxide ion?
 (A) $\text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$ (B) $\text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (C) $\text{O}_2 < \text{O}_2^- > \text{O}_2^{2-}$ (D) $\text{O}_2 > \text{O}_2^- < \text{O}_2^{2-}$
16. Which of the following is paramagnetic in nature?
 (A) C_2^{2-} (B) N_2^{2+}
 (C) O_2^+ (D) B_2^{2+}

Comprehension 2:

If the electrons in the atoms are filled by following rule.

- (1) Aufbau principle :- The electrons are added one by one into various orbital in order of their increasing energy.
- (2) Pauli's exclusion principle :- No two electrons can have same set of all quantum number.
- (3) Hund's rule of minimum multiplicity :- Opposite spin of electrons in alternate degenerate orbitals leads for minimum energy configuration then

17. The electronic configuration of Cr(24) and Cu(29) respectively is -
 (A) $[\text{Ar}] 4s^2 3d^4$, $[\text{Ar}] 4s^2 3d^9$ (B) $[\text{Ar}] 4s^2 3d^5$, $[\text{Ar}] 4s^1 3d^{10}$
 (C) $[\text{Ar}] 4s^1 3d^5$, $[\text{Ar}] 4s^2 3d^9$ (D) $[\text{Ar}] 4s^2 3d^4$, $[\text{Ar}] 4s^1 3d^{10}$
18. Magnetic moment (spin only) of Fe^{+2} is –
 (A) $\sqrt{3}$ (B) $\sqrt{8}$
 (C) $\sqrt{35}$ (D) $\sqrt{48}$

PART – B**(Matrix Match Type)**

This section contains **2 questions**. Each question has two columns. For each statement in Column I, there may be **ONE or MORE THAN ONE** correct choice(s) in Column II.

1. Match Column-I (atomic number of elements) with Column-II (position of elements in periodic table) and select the correct answer using the codes given below the lists:

Column I		Column II	
(A)	19	(p)	p-block
(B)	22	(q)	f-block
(C)	32	(r)	d-block
(D)	64	(s)	s-block

2. Match Column I with Column II:

Column I		Column II	
(A)	PCl_5	(p)	Two axial bonds + three lone pairs
(B)	XeF_2	(q)	Two axial + two equatorial bonds
(C)	SF_4	(r)	Two axial + one equatorial bonds
(D)	ClF_3	(s)	Three equatorial + two axial bonds

Section – III (Mathematics)

PART – A

(Single Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Value of $1 - \cot 23^\circ =$

(A) $\frac{2}{1 - \cot 23^\circ}$

(C) $\frac{2}{1 - \cot 22^\circ}$

(B) $\frac{4}{1 - \cot 22^\circ}$

(D) none of these
2. If $\sin 2\theta = \cos 3\theta$ and θ is an acute angle, then $\sin \theta$ equals

(A) $\frac{\sqrt{5}-1}{4}$

(C) $\frac{\sqrt{5}+1}{4}$

(B) $-\left(\frac{\sqrt{5}-1}{4}\right)$

(D) $\frac{-\sqrt{5}-1}{4}$
3. The value of $\sin 6^\circ \cdot \sin 30^\circ \cdot \sin 54^\circ \cdot \sin 66^\circ$ is

(A) $\frac{\sqrt{5}-1}{32}$

(C) $\frac{\sqrt{5}+2}{8}$

(B) $\frac{\sqrt{5}-1}{8}$

(D) $\frac{\sqrt{5}+2}{32}$
4. Value of x such that $|x-3| \in [1, 3]$ and $x^2 - 3x - 4 \leq 0$

(A) $[0, 2] \cup \{4\}$

(C) $[0, 2]$

(B) $[-1, 4] - \{2\}$

(D) $[2, 4]$
5. If the line $y = \sqrt{3}x$ cuts the curve $x^3 + y^2 + 3x^2 + 9 = 0$ at the points A, B, C, then OA.OB.OC (O being origin) equals

(A) 12

(C) -12

(B) 72

(D) -72
6. If pairs of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angles between the other pair, then

(A) $pq = 1$

(C) $pq = 2$

(B) $pq = -1$

(D) $pq = -1/2$
7. A straight Line L through the point $(3, -2)$ is inclined at an angle of 60° to line $\sqrt{3}x + y = 1$. It L also intersects the x-axis then the equation of L is

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

8. If $(1, 2)$, $(-1, 3)$ are on the same side of the line $3x - 4y = a$, then a is
- (A) $-8 < a < -5$ (B) $a = -8$
 (C) $a = -15$ (D) $a < -15$ or $a > -5$

PART – A

(Multi Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

9. The point of intersection of lines $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$ lies on the line
- (A) $x - y = 0$ (B) $(x + y)(a + b) = 2ab$
 (C) $(lx + my)(a + b) = (l + m)ab$ (D) $(lx + my)(a + b) = (l - m)ab$
10. $\sin^4 x + \cos^4 x = a$ has a real solution for
- (A) $a \in \mathbb{R}$ (B) $a = \frac{1}{2}$ (C) $a = \frac{3}{2}$ (D) $a = \frac{1}{4}$
11. The radius of the circle which touches the co-ordinate axes and the line $3x + 4y = 12$ is
- (A) 1 (B) 2 (C) 4 (D) 6
12. The equation of straight line equally inclined to the axes and equidistant from the point $(1, -2)$ and $(3, 4)$ is
- (A) $x + y = 3$ (B) $x - y + 1 = 0$ (C) $y - x = 2$ (D) $y - x + 1 = 0$
13. If $2\cos x + \sin x = 1$, then value of $7\cos x + 6\sin x$ can be
- (A) 2 (B) 6 (C) 3 (D) 1
14. Let $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$ then
- (A) $P(3) = \frac{1}{16}$ (B) $P(4) = \frac{2 - \sqrt{2}}{16}$ (C) $P(5) = \frac{3 - \sqrt{5}}{32}$ (D) $P(6) = \frac{2 + \sqrt{3}}{16}$

PART – A

(Comprehension Type)

This section contains **2 Paragraphs**. Based upon each paragraph, there are **2 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Comprehension 1:

A variable circle of radius 2 units rolls outside the circle $x^2 + y^2 + 4x = 0$. If c and c_1 denotes the centre of respective circles, then

15. Locus of centre of variable circle i.e. c is
- (A) $x^2 + y^2 + 4x = 0$ (B) $x^2 + y^2 + 4x - 12 = 0$
 (C) $x^2 + y^2 + 4x - 12 = 0$ (D) none of these
16. If the line joining c and c_1 makes an angle of 60° with x -axis, then equation of common tangents of the circles at their point of contact is
- (A) $x + \sqrt{3}y - 2 = 0$, $x + \sqrt{3}y + 6 = 0$ (B) $x - \sqrt{3}y - 2 = 0$, $x - \sqrt{3}y + 6 = 0$
 (C) $x + \sqrt{3}y + 2 = 0$, $x + \sqrt{3}y - 6 = 0$ (D) $x - \sqrt{3}y + 2 = 0$, $x - \sqrt{3}y - 6 = 0$

Comprehension 2:

Consider a variable line 'L' which passes through the point of intersection 'P' of the lines $3x + 4y - 12 = 0$ and $x + 2y - 5 = 0$ meeting the coordinate axes at points A and B.

17. Locus of the middle point of the segment AB has the equation
 (A) $3x + 4y = 4xy$ (B) $3x + 4y = 3xy$ (C) $4x + 3y = 4xy$ (D) $4x + 3y = 3xy$
18. Locus of the feet of the perpendicular from the origin on the variable line 'L' has the equation
 (A) $2(x^2 + y^2) - 3x - 4y = 0$ (B) $2(x^2 + y^2) - 4x - 3y = 0$
 (C) $x^2 + y^2 - 2x - y = 0$ (D) $x^2 + y^2 - x - 2y = 0$

PART – B**(Matrix Match Type)**

This section contains **2 questions**. Each question has two columns. For each statement in Column I, there may be **ONE or MORE THAN ONE** correct choice(s) in Column II.

- | | | |
|----|--|---|
| 1. | Column-1
(A) Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is
(B) A chord of the circle $(x-1)^2 + y^2 = 4$ lies along the line $y = 2\sqrt{3}(x-1)$. The length of the chord is equal to
(C) The number of circles touching all the three lines $3x + 7y = 2$, $21x + 49y = 5$ and $9x + 21y = 1$ are
(D) If radii of the smallest and largest circle passing through the point $(\sqrt{3}, \sqrt{2})$ and touching the circle $x^2 + y^2 - 2\sqrt{2}y - 2 = 0$ are r_1 and r_2 respectively, then the mean of r_1 and r_2 is | Column-2
(p) 4

(q) 2

(r) 0

(s) 1 |
| 2. | Find the value of λ if the family of straight lines $(2x + 3y + 4) + \lambda(6x - y + 12) = 0$ is
Column-I
(A) $\parallel$ to y-axis
(B) $\perp$ to $7x + y - 4 = 0$
(C) Passes through (1, 2)
(D) $\parallel$ to x-axis | Column-II
(p) $\lambda = -3/4$
(q) $\lambda = -1/3$
(r) $\lambda = -17/41$
(s) $\lambda = 3$ |

FIITJEE INTERNAL TEST

Batch: LKTYR235A03

IIT - JEE 2025

PHASE – I

PAPER – I

ANSWERS

QP CODE:

Physics

Part – A

- | | | | |
|--------|--------|---------|--------|
| 1. C | 2. B | 3. A | 4. B |
| 5. B | 6. C | 7. B | 8. C |
| 9. AB | 10. AD | 11. ABD | 12. AC |
| 13. CD | 14. AD | 15. D | 16. A |
| 17. A | 18. B | | |

Part – B

- (A) – (r); (B) – (s); (C) – (p); (D) – (q)
- (A) – (s); (B) – (q); (C) – (r); (D) – (p)

Chemistry

Part – A

- | | | | |
|--------|--------|----------|--------|
| 1. A | 2. B | 3. C | 4. B |
| 5. C | 6. D | 7. D | 8. C |
| 9. ABC | 10. BD | 11. ABCD | 12. AC |
| 13. AD | 14. CD | 15. B | 16. C |
| 17. D | 18. C | | |

Part – B

- (A) – (s); (B) – (r); (C) – (p); (D) – (q)
- (A) – (s); (B) – (p); (C) – (q); (D) – (r)

Mathematics

Part – A

- | | | | |
|--------|---------|---------|--------|
| 1. C | 2. A | 3. B | 4. A |
| 5. B | 6. A | 7. B | 8. D |
| 9. ABC | 10. B | 11. ABD | 12. AD |
| 13. AB | 14. ABC | 15. B | 16. A |
| 17. A | 18. B | | |

Part – B

- (A) – (q); (B) – (p); (C) – (r); (D) – (s)
- (A) – (s); (B) – (r); (C) – (p); (D) – (q)